

Unit 6, Lesson 3: Exploring Rates

Benchmark

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

Learning Targets

- ☐ I can use ratios and rates to describe relationships between quantities.
- ☐ I can write a ratio as a unit rate.

6.3.1 Warm-Up: Would You Rather?

1. Would you rather share the small bag of chips with 1 friend or the large bag of chips with 8 friends? Explain your reasoning using mathematics.

1.5 ounces



8 ounces

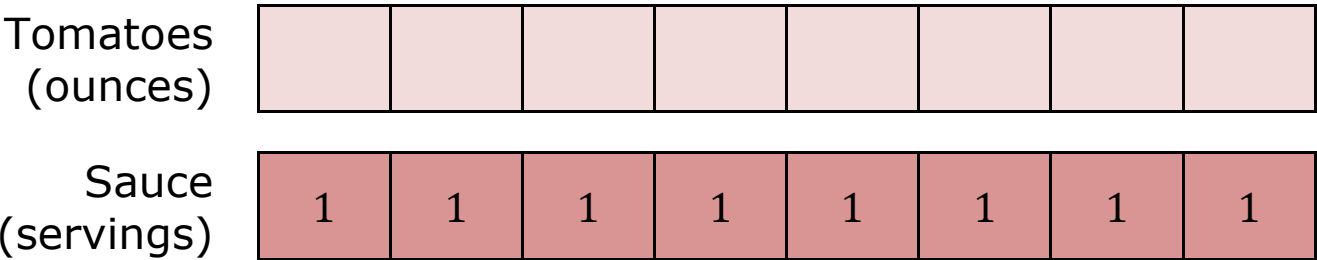


6.3.2 Collaboration: Expressing Ratios in Different Ways

Ratios are often expressed in the real world as an amount of “something per something.” For example, in setting the world record, Joey Chestnut ate 7.4 hot dogs per minute.

Discuss each of the following scenarios with your partner and work together to answer the questions.

1. A recipe for marinara sauce requires 56 ounces (oz) of whole peeled tomatoes. The recipe yields 8 servings of marinara sauce.
- a. What is the ratio of tomatoes to sauce?
- b. Complete the tape diagram to represent the ratio of tomatoes to sauce.



- c. How many ounces of whole peeled tomatoes are in 1 serving of marinara sauce?
- d. Complete the ratio.

oz of tomatoes

1 serving

2. In 2012, there were 982 farms in Hardee County, Florida. The total amount of farmland was 273,978 acres.

a. Write the ratio of acres : farms.

b. Complete the statement.

The total amount of farmland in Hardee County is split up among all the farms, so to determine the average number of acres for 1 farm,

- add 982 to 273,978.
- subtract 982 from 273,978.
- multiply 273,978 by 982.
- divide 273,978 by 982.

c. Complete the ratio.

$$\frac{\underline{\hspace{1cm}} \text{ acres}}{1 \text{ farm}}$$

d. Discuss with your partner whether you think this number of acres per 1 farm means that every farm in Hardee County was that size in 2012. Then, write a brief summary of your reasoning.

6.3.3 Guided Instruction: Understanding Rates

Each of the ratios discussed so far in this lesson are also **rates**.



A **rate** is a ratio that compares two quantities of different units.

A ratio is a comparison of two numbers, so all rates can also be expressed using ratios. Rates are often used to express how long it takes to do something.

1. Consider a familiar scenario involving a rate.

Tyler ran 2 miles in 30 minutes.

- a. Draw a tape diagram for the ratio that represents Tyler's run.

- b. Complete the ratios. Then, complete each statement to interpret the ratios.

$$\frac{\underline{\hspace{1cm}} \text{ miles}}{1 \text{ minute}}$$

$$\frac{\underline{\hspace{1cm}} \text{ minutes}}{1 \text{ mile}}$$

Tyler runs _____ miles for every 1 minute.

Tyler runs _____ minutes for every 1 mile.

- c. Discuss with your partner how to use one of the rates to determine how many miles Tyler runs in an hour. Then, complete the statement.

Tyler runs _____ miles per hour.

Each of the rates used to describe Tyler's run is a **unit rate**.



A **unit rate** is a ratio comparing a number of units of one quantity to 1 unit of a second quantity.

6.3.4 Guided Practice: Using Ratios and Rates

1. A nutritional label for raspberry ice cream is shown.

a. What is the ratio of calories to servings?

b. What is the ratio of total carbohydrates in the container to the number of calories in the container?

c. What is the ratio of added sugars to the number of servings?

d. Complete the unit rate to express the number of added sugars per serving.

$$\frac{\underline{\hspace{1cm}} \text{ grams}}{1 \text{ serving}}$$

Nutrition Facts		
3 servings per container		
Serving size		2/3 Cup (123g)
Calories	Per Serving	
	330	
	% Daily Value*	
Total Fat	19g	24%
Saturated Fat	12g	60%
Trans Fat	0.5g	
Cholesterol	130mg	43%
Sodium	180mg	8%
Total Carbohydrate	35g	13%
Dietary Fiber	0g	0%
Total Sugars	29g	
Includes Added Sugars	24g	48%
Protein	5g	
Vitamin D	1mcg	6%
Calcium	135mg	10%
Iron	1mg	6%
Potassium	177mg	4%

*The % Daily Value tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.

6.3.5 Your Turn!

1. Tracy McGrady is a former NBA basketball player from Bartow, Florida. He played in 938 games in the NBA, scoring a total of 18,381 points. Complete the ratio to express the number of points he scored per game as a unit rate. Round the number of points to the nearest tenth.

$$\frac{\underline{\hspace{1cm}} \text{ points}}{1 \text{ game}}$$

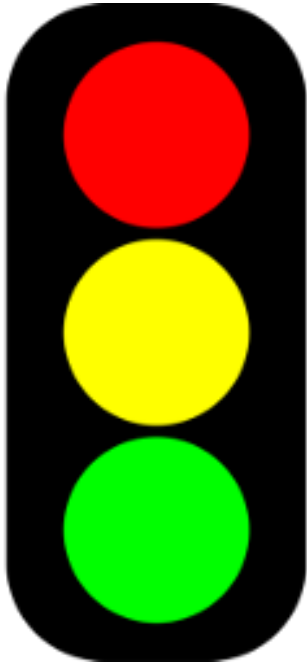
2. The ruby-throated hummingbird is the most common hummingbird found in Florida. This bird can flap its wings 1,590 times in 30 seconds. Complete the ratio to express how fast the ruby-throated hummingbird flaps its wings per second.

___ wing flaps : 1 second



6.3.6 Wrap-Up: Reflection

1. Complete the stoplight reflection to share how you are feeling about the concepts learned in today's lesson.



I still feel stuck about how to ...

I'm starting to get the hang of ...

I feel confident and ready to go with ...